



FLOW & ARCHITECTURE DIAGRAMS

AI-Powered Contract Intelligence & Clause Extraction Platform

Supporting Visual Reference for the C.L.E.A.R. BRD (v1.1) and SRS (v1.1)

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Legal & Secretarial	Diagram Pack	1.2	01 Aug 2026

WHAT THIS DOCUMENT CONTAINS

Three diagrams supporting the C.L.E.A.R. requirements documents: (1) the document processing flow, from intake through ADI parsing, LLM classification, header-based or chunk-based clause extraction, and persistence into Doc Master and Doc Content Master; (2) the Contract Snapshot and AI Insights dashboard fields; and (3) the semantic query (RAG) pipeline used to answer natural-language questions over the stored contracts.

1. Document Processing Flow

This diagram traces a single contract from submission through to persistence. After ADI-based OCR/parsing produces a structured JSON with section headings and title, an LLM identifies the document type. The system then checks whether a usable section header or title exists: if yes, an LLM identifies the clause based on document type for that section; if no, the document is split into 4-page chunks to locate clauses instead. Once a clause is found, it is mapped to the Clause Master, and results are persisted into two tables — Doc Master (DocId, Path, DocType, JSON path) and Doc Content Master (DocId, Clauses, Reference Page, Embedding, Content) — which together feed the Contract Snapshot and AI Insights dashboard.

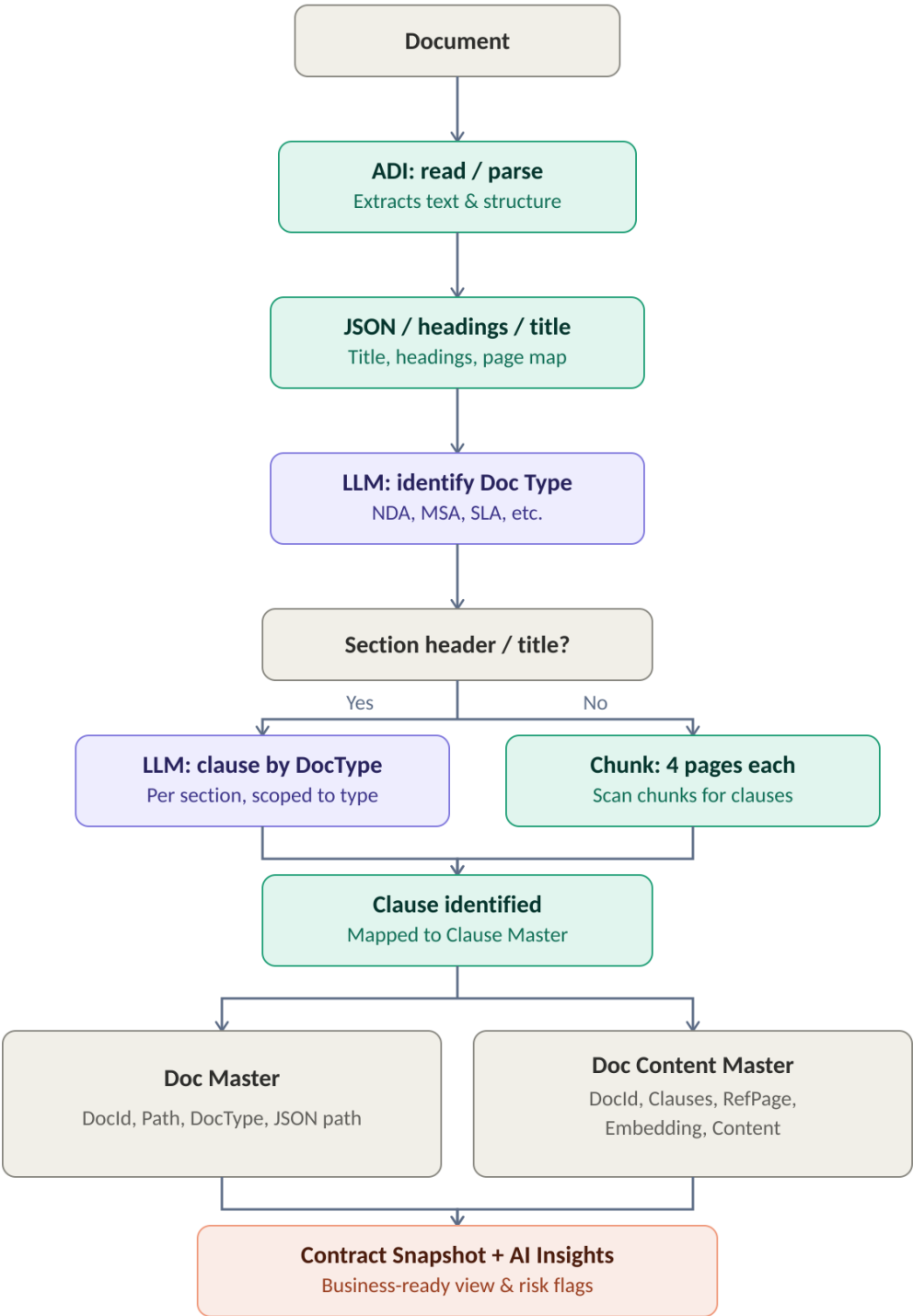


Figure 1 — C.L.E.A.R. document processing flow, from intake to Doc Master / Doc Content Master persistence.

REFERENCE

This flow corresponds to BRD Section 7.1 (Backend Flow) and Section 8 (Data Model), and SRS Section 5 (AI Processing Flow) and Section 7 (Data Requirements).

2. Contract Snapshot & AI Insights Dashboard

Once clauses are persisted, the dashboard groups them into three field sets. Contract information covers type, value, payment terms, auto renewal and governing law. Important dates cover effective date, expiry date and term of contract. Clause summary flags high-risk clauses and missing clauses. Below these, the AI Insights panel surfaces risk, renewal, business and review insights drawn across the repository.

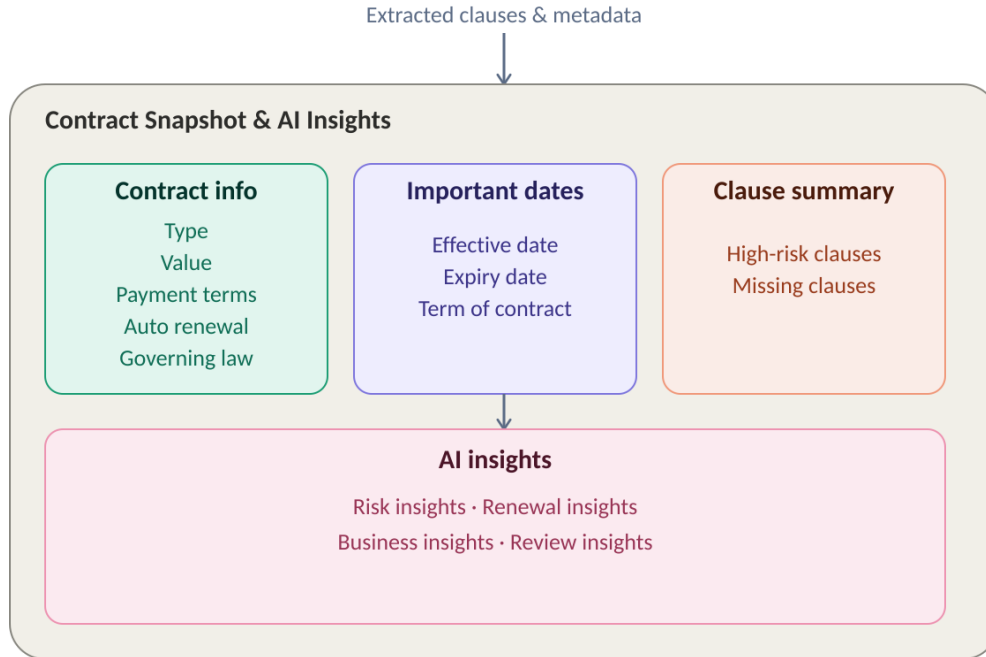


Figure 2 — Contract Snapshot field groups and AI Insights categories.

REFERENCE

This corresponds to BRD Section 9 (Dashboard Requirements) and SRS Section 6 (Dashboard Requirements), and functional requirements FR-DASH-01 and FR-DASH-02.

3. Semantic Query (RAG) Pipeline

This diagram shows how a user's natural-language question is answered. A LangChain pipeline first runs query analysis with an LLM to determine intent, likely document type and a confidence score, then generates a query embedding. That embedding drives a vector search over Postgres with pgvector, filtered by ProjectId, ContractId and, optionally, DocumentTypeId. The top-K matches are retrieved, optionally reranked, assembled into a prompt, and passed to an LLM to generate the final answer, which is returned to the user together with its supporting sources.

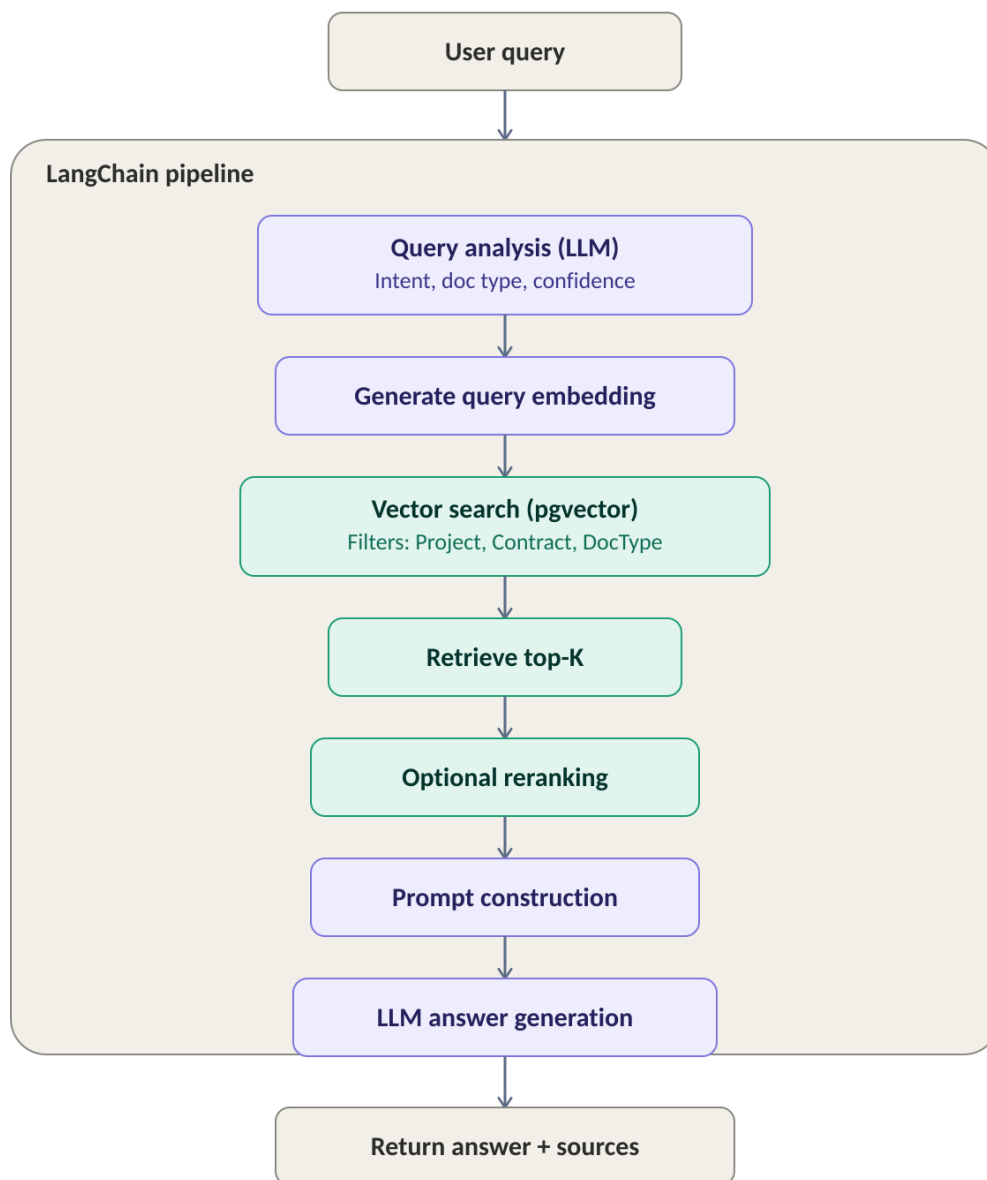


Figure 3 — Semantic query pipeline from user question to answer with sources.

REFERENCE

This supports the semantic search requirement in BRD Section 3.1 (AI Processing) and SRS FR-SEARCH-01, and extends it with the retrieval-augmented generation flow used for natural-language question answering (BRD Section 15.2 / SRS FAC-02).